

SUBJECT: APPLICATIONS DEVELOPMENT THEORY (APT470S) Full-time and part-time	EXAM DATE: November 2024	DURATION: 2.5 Hours	TOTAL MARKS: 60
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**Cape Peninsula
University of Technology**
creating futures

**DEPARTMENT OF INFORMATION TECHNOLOGY
FACULTY OF INFORMATICS AND DESIGN**

COURSE(S): **ADICTA: ADVANCED DIPLOMA IN INFORMATION AND COMMUNICATION
TECHNOLOGY: APPLICATIONS DEVELOPMENT**

EXAMINER: **MS G KHAN**
MODERATOR: **MR T MAKHURANE**
EXTERNAL MODERATOR: **DR M NORMAN**

GENERAL INSTRUCTIONS

- You are required to answer all questions.
- Read the questions carefully before answering them.
- This test **counts 30%** of your final mark for this subject and covers work from terms 3 and 4 as per the topic list in the Subject Guide.

QUESTION 1**(20 MARKS)**

You are the chief developer at a software company that considers quality as of the utmost importance in both the products that it produces and the processes that it uses to achieve it. At present, several quality-related activities are planned to assist the software development process. Based on this answer the following questions.		
1.1	<i>"An effective software process, applied in a manner that creates a useful product that provides measurable value for those who produce it and those who use it."</i> Consider the definition for software quality provided above then explain what is meant by each of the following terms – <i>a) An effective software process</i> <i>b) a useful product</i> <i>c) value for those who produce it and</i> <i>d) those who use it</i>	10
1.2	You wish to emphasize the importance of having a testing strategy during quality assurance. Provide the key reasons that you would include in your explanation.	4
1.3	You have decided that desk checking does not always have to be a formal process. Provide three examples of instances when this could be the case and then suggest ways in which you can improve the efficacy of such reviews.	6

QUESTION 2**(20 MARKS)**

2.1	You are working on the security plan of an app for a client that owns several pharmacies and wellness clinics across the country. This is a huge task that needs careful thought and solutions. Explain the security considerations and precautions that you will include in the mobileApp.	10
2.2	Who should conduct the security testing of the mobileApp for the pharmacy client? Why?	5
2.3	Assume that a new function has been implemented in the mobileApp to check for food allergens in products sold at the pharmacy and wellness centres. Explain how you - as a release tester - would test this functionality.	5

QUESTION 3**(20 MARKS)**

3.1	Effective software project management is influenced by the <i>people, products, and processes</i> employed. Explain how these factors affect the management of software projects.	6
3.2	To manage the various changes that will be made to work you decide to employ good Software Configuration Management (SCM) practices. Identify the elements in the SCM repository that would have to change during work activities and provide a sample document associated with each element.	10
3.3	A risk analysis is being conducted to understand the level of uncertainty and the degree of loss that such risks pose to the project. Discuss the risks that could threaten the success of a project.	4

***** END OF PAPER *****